A Determinant-Sign Case of the Four-Dimensional Negative Goldberg Problem

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Abstract

A closed negative Einstein almost-Kähler four-manifold is Kähler if $\det(W^+ + s\,\mathrm{Id}\,/12) \le 0$ everywhere in its almost-complex orientation. The proof combines the classical Einstein Weyl–Weitzenböck equation with Armstrong's existence of a torsion zero. A scalar maximum principle first forces the shifted curvature operator to be nonpositive. The torsion zero then places a second elementary symmetric function on the boundary of its positive cone; the strong maximum principle forces a parallel rank-one operator and finally the parallelness of the given fundamental form. The determinant assumption is additional: the general negative Goldberg problem remains the case not covered by this sign condition.

1 Statement and normalization

All manifolds below are smooth, closed (compact without boundary), connected, and four-dimensional. Curvature on two-forms is normalized by

$$\mathcal{R} = \begin{pmatrix} W^+ + s\,\mathrm{Id}\,/12 & B^* \\ B & W^- + s\,\mathrm{Id}\,/12 \end{pmatrix}.$$

Weyl norms are Hilbert–Schmidt norms of endomorphisms. The scalar and rough Laplacians are nonnegative: $\Delta = -\operatorname{div}\nabla$ and $\nabla^*\nabla$. Set

$$\mathrm{Ric} = -ag, \qquad a > 0, \qquad s = -4a, \qquad \mathcal{R}_+ = W^+ - \frac{a}{3}\,\mathrm{Id}\,.$$

Theorem 1.1. *Let (M, g, J, Ω) be almost-Kähler and Einstein with $\mathrm{Ric} = -ag$, $a > 0$. Choose the orientation induced by J. If*

$$\det \mathcal{R}_+ \le 0 \quad \textit{everywhere on } M,$$

then $\nabla\Omega = 0$. In fact, for $n = \Omega/\sqrt{2}$,

$$\mathcal{R}_+ = -a\, n \otimes n.$$

The hypothesis involves the full self-dual curvature block, not merely W^+, not the determinant of the full six-dimensional curvature operator, and not sectional-curvature negativity. The proof below uses neither project FD/N1 estimates nor a uniform compactness assumption over a family of metrics. The inputs are classical identities and Armstrong's topological observation; no literature priority is asserted for their composition here.

2 The scalar equation that supplies the sign

Put

$$A = -\mathcal{R}_+ = \frac{a}{3}\,\mathrm{Id} - W^+, \qquad \operatorname{tr} A = a, \qquad u = \sigma_2(A) = \frac{a^2 - |A|^2}{2} = \frac{a^2}{3} - \frac{1}{2}|W^+|^2.$$

These definitions are smooth even at repeated curvature eigenvalues.

Lemma 2.1. *For every closed oriented Einstein four-manifold with* $\mathrm{Ric} = -ag$*, one has*

$$\boxed{(\Delta + 2a)u = |\nabla A|^2 + 18 \det A.} \tag{1}$$

Proof. Write $W = W^+$. The classical Einstein Weyl–Weitzenböck equation is

$$\nabla^*\nabla W + \frac{s}{2}W - 6W^2 + 2|W|^2\,\mathrm{Id} = 0. \tag{2}$$

See Derdziński [1]; the same operator convention is written explicitly in [2, Lemma 2.5, $f = 1$]. Since $\operatorname{tr} W = 0$ and $\operatorname{tr}(W^3) = 3 \det W$,

$$\Delta|W|^2 = -s|W|^2 + 36 \det W - 2|\nabla W|^2.$$

The shift $W = a\,\mathrm{Id}\,/3 - A$ gives

$$|W|^2 = \frac{2a^2}{3} - 2u, \qquad \det W = -\frac{2a^3}{27} + \frac{a}{3}u - \det A.$$

As $\Delta u = -\frac{1}{2}\Delta|W|^2$, substitution gives $\Delta u = -2au + 18 \det A + |\nabla A|^2$, as asserted. □

Lemma 2.2. *If* $\det \mathcal{R}_+ \le 0$ *everywhere, then* $A \succeq 0$ *everywhere. If, in addition,* $u(p) = 0$ *at any point, then*

$$u \equiv 0, \qquad \nabla A = 0, \qquad A^2 = aA, \qquad \operatorname{rank} A = 1.$$

Proof. Since the block has odd rank, the determinant assumption is precisely $\det A \ge 0$. Equation (1) implies $(\Delta + 2a)u \ge 0$. A negative minimum of u would satisfy $\Delta u \le 0$ and $2au < 0$, a contradiction. Hence $u \ge 0$.

We use the following elementary fact: a real symmetric 3×3 matrix with positive trace, nonnegative σ_2, and nonnegative determinant is positive semidefinite. If there is exactly one negative eigenvalue and the other two are positive, the determinant is negative. If they are $-x, -y, a + x + y$ with $x, y > 0$, then

$$\sigma_2 = xy - (x + y)(a + x + y) < 0.$$

If one eigenvalue is zero, a negative eigenvalue and positive trace likewise force $\sigma_2 < 0$. The remaining cases are incompatible with positive trace. Applying this fact proves $A \succeq 0$.

If $u(p) = 0$, the strong minimum principle, applied to the nonnegative function u with $(\Delta + 2a)u \ge 0$, gives $u \equiv 0$. Equation (1) then gives $\nabla A = 0$ and $\det A = 0$. A positive semidefinite matrix of trace a and $\sigma_2 = 0$ has eigenvalues $(a, 0, 0)$, proving the remaining assertions. □

Thus, on this closed negative Einstein class, the global determinant-sign condition is equivalent to $\mathcal{R}_+ \preceq 0$. This equivalence uses the Einstein equation and the maximum principle; it is false for arbitrary pointwise symmetric matrices with negative trace.

3 A torsion zero forces the rigid boundary case

We recall the classical global input in the precise form needed.

Lemma 3.1 (Armstrong)**.** *A closed Einstein almost-Kähler four-manifold with $s \neq 0$ has a point where $\nabla\Omega = 0$.*

Proof. The Nijenhuis tensor is a section of the complex rank-two bundle

$$\mathcal{N} = \Lambda^{0,2}T^*M \otimes T^{1,0}M.$$

Its second Chern number is

$$c_2(\mathcal{N})[M] = 2c_1(TM, J)^2 + c_2(TM, J) = 5\chi + 6\tau.$$

If it were nowhere zero, this number would vanish. On an Einstein four-manifold with nonzero scalar curvature, Chern–Weil gives

$$8\pi^2\chi = \int \left(|W^+|^2 + |W^-|^2 + \frac{s^2}{24} \right) > 0,$$

$$4\pi^2(2\chi + 3\tau) = \int \left(2|W^+|^2 + \frac{s^2}{24} \right) > 0.$$

Consequently $5\chi + 6\tau = \chi + 2(2\chi + 3\tau) > 0$, a contradiction. On an almost-Kähler manifold, vanishing of the Nijenhuis tensor at a point is equivalent to vanishing of $\nabla\Omega$ there. This is [3, Theorem 2.2.4], whose topological proof is reproduced here. □

Proof of Theorem 1.1. Let $n = \Omega/\sqrt{2}$. It is a harmonic self-dual two-form of unit length. The harmonic-form Weitzenböck equation gives

$$\nabla^*\nabla n = 2W^+ n - \frac{s}{3}n = 2(a\,\mathrm{Id} - A)n. \tag{3}$$

Since $|n| = 1$, putting $q = |\nabla\Omega|^2 = 2|\nabla n|^2$ yields

$$q = 4\big(a - \langle An, n\rangle\big). \tag{4}$$

Choose p from Lemma 3.1. Lemma 2.2 gives $A \succeq 0$; equation (4) gives $\langle A(p)n(p), n(p)\rangle = a = \operatorname{tr} A(p)$. It follows that

$$A(p) = a\, n(p) \otimes n(p), \qquad u(p) = 0.$$

The second part of Lemma 2.2 now applies. In particular, $\Pi = A/a$ is a globally defined parallel orthogonal rank-one projection.

Define $v = \Pi n$ without making any orientation choice for the image line bundle. By (3) and $A^2 = aA$,

$$\nabla^*\nabla v = \Pi\nabla^*\nabla n = 2\Pi(a\,\mathrm{Id} - A)n = 0.$$

Integration over the closed manifold gives $\nabla v = 0$. At p, $v(p) = n(p)$, so $|v| \equiv 1$. Since $|n| \equiv 1$ and Π is an orthogonal projection, $n = v$ everywhere. Therefore $\nabla n = 0$, hence $\nabla\Omega = 0$. The relation $\mathcal{R}_+ = -a\, n \otimes n$ follows as well. □

4 Actual consequences and the remaining case

Corollary 4.1. *Every closed strictly almost-Kähler negative Einstein four-manifold would have an open set on which* $\det \mathcal{R}_+ > 0$*. On that set the signature of* $\mathcal{R}_+$ *is* $(+,-,-)$*.*

Proof. The first statement is the contrapositive of the theorem and continuity. The determinant is positive, while $\operatorname{tr} \mathcal{R}_+ = -a < 0$, so exactly one eigenvalue is positive and the other two are negative. □

Corollary 4.2. *A closed negative Einstein four-manifold with* $\mathcal{R}_+ \prec 0$ *everywhere has no compatible almost-Kähler structure in the stated orientation.*

Proof. Such a structure would imply $\mathcal{R}_+ = -a\, n \otimes n$, which has two zero eigenvalues. Alternatively, strict negativity makes the right side of (4) positive everywhere, contradicting Lemma 3.1. □

Fine–Premoselli construct compact, negatively curved, non-locally-homogeneous Einstein four-metrics and explicitly state that their examples satisfy $W^+ + s\,\mathrm{Id}\,/12 \prec 0$ [4, Theorem 1.2 and following discussion]. Corollary 4.2 applies to those metrics in that orientation. This concerns compatibility with the specified metrics, not nonexistence of symplectic forms or Einstein metrics on the underlying smooth manifolds. No claim about the opposite orientation is needed here.

In the preceding harmonic-product representation, this says that under the determinant condition every positive trace-one rank-one element of the harmonic multiplication kernel comes from a parallel form. In the strictly negative-definite case there is no such rank-one element at all. The Einstein PDE, rather than positivity of the finite-dimensional matrices alone, supplies this conclusion.

Scope. The original Q4 hypotheses do not include $\det \mathcal{R}_+ \leq 0$. No derivation of that sign from those hypotheses is given here. The proof settles this specified curvature-sign case and identifies a necessary open indefinite-curvature region in any hypothetical strict closed example. It neither proves full negative Goldberg nor supplies a closed counterexample.

References

[1] A. Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*, Compositio Math. **49** (1983), 405–433. `https://archive.numdam.org/item/CM_1983__49_3_405_0/`.

[2] G. Catino and D. Dameno, *Conformal Kähler Rigidity of Einstein Four-Manifolds*, arXiv:2607.21538v1 (2026), Lemma 2.5. `https://arxiv.org/html/2607.21538v1`. Only the classical Weyl–Weitzenböck identity is used.

[3] J. Armstrong, *Almost-Kähler Geometry*, D.Phil. thesis, Oxford (1998), Theorem 2.2.4. `https://johnarmstrongmaths.com/publications/resources/thesis.pdf`.

[4] J. Fine and B. Premoselli, *Examples of compact Einstein four-manifolds with negative curvature*, arXiv:1802.00608, Theorem 1.2 and the curvature inequality following it. `https://arxiv.org/html/1802.00608`.